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** Please note that our demo is accessible by downloading our attached files or the files linked on the github page below and running “python main.py” in your terminal. This should open a website locally. We were trying to upload our work onto our github.io page, but were running into trouble and were unable to figure it out before the deadline. We plan to have it available on the github page for our final submission. **

https://github.com/letmebyourii/letmebyourii.github.io

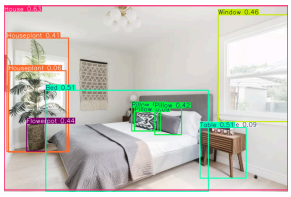
Dataset:

We initially planned to use Mask R-CNN on the COCO dataset, but in discussing our project with the TA, we began exploring Ultralytics YOLO models. Unfortunately, when attempting to finetune pre-trained YOLO models on additional datasets (such as the Objects365 Dataset, which additionally is supported by YOLO through YAML files), we consistently ran into storage issues due to hardware. For this reason, we intend to rely on pretrained models until we can figure out a less hardware-demanding approach to train on more data. In building our familiarity with computer vision, we experimented with various YOLO models on both an image of a bedroom and a video from an HGTV show focusing on furniture and rooms. There is more detail about these models and their datasets within the “Background Section” of our project, and the following is simply a summary of our experiment so far.

We first tested the YOLO26 and YOLO26-seg models, both trained on the COCO dataset. These models have various versions based on increasing performance values. A potential benefit of using a segmentation model is having the distinct shapes of items being identified, allowing for more distinct navigation instructions. We additionally tested the YOLOv8n model, with two variants trained on the COCO dataset and the Open Images V7 dataset, as well as the YOLOv8s-worldv2 model.

With conf=0.05:

Model	**Dataset**	**Image**	**Video**
yolo26n	COCO		https://youtu.be/MNIUXqVNBss

yolov8n	COCO		https://youtu.be/WdUvxKlddZ8
yolov8n-oiv7	Open Images V7		https://youtu.be/XUGolmxVkEo
yolo8s-worldv2	COCO		https://youtu.be/rC8zAEQPmoY
yolo26n-seg	COCO		https://youtu.be/vayq5U79ou4

Our current model choice is the YOLOv8n model trained on the Open Images V7 dataset ("yolov8n-oiv7") due to the large number of classes covered by the model and its performance on our sample image compared to the other models. We also will keep in mind that the video that we tested each model on is not necessarily representative of the type of video that our model will operate on, as the movement in the video sample is very fast. A slower video pace may mean that we can look into larger models within YOLOv8 trained on Open Images V7.

We also tried to train these models on the Objects365 dataset, but we ran into storage errors again. Now that we have a better understanding of the baseline models, we hope we can find another dataset to finetune this model. This is our planned next step, and we plan to look into using the datasets used within MontageNet listed under the "Related Work" section to find a dataset suitable for fine tuning, such as the NYUD-V2 dataset that includes information on depth that may be especially useful for navigation through a room (Iuan Kai Fang et al. 2023).

Initial ML pipeline:

A user will open up our website on their phone, which will access their back camera. The website will be running the ML model on the user's livestream video. The user will verbally

indicate which object within the frame they would like to walk to. The website will verbally dictate how to navigate around other objects within the frame in order to reach the goal object.

Technical blog post:

As Machine Learning (ML) becomes increasingly popular, ML applications can offer accessible ways to interact with the world. Computer Vision (CV), a sect of deep learning and ML, introduces approaches to process visual data, including with the detection and segmentation of objects (Voulodimos et al. (2018)).

In particular, computer vision has emerged as a powerful tool for addressing challenges faced by visually impaired individuals. According to the World Health Organization, approximately 285 million people worldwide live with blindness or severe visual impairment ((Kadhim and Oleiwi (2020))). The reality of these individuals pushed researchers to develop assistive technologies like the YOLO (You Only Look Once) algorithm. YOLO processes an input image through a series of convolution neural networks (CNN) to produce bounding boxes that identify and locate objects within a specific environment (Kadhim and Oleiwi (2020)). The algorithm gained recognition for aiding visually impaired people with fast, accurate, realtime navigation.

Our project aims to connect a livestream to the YOLO algorithm to guide a user toward a specific object. While this concept is not entirely new, it continues to evolve as newer and more advanced versions of the YOLO algorithm are developed. A study conducted by Changchun University found that the YOLO model performs poorly when detecting overlapping objects, small objects, or objects under varying lighting conditions. To address these challenges, the researchers added three additional components: a Feature Weighting Block (FWB), an Adaptive Bottleneck Block (ABB), and an Enhanced Feature Attention Head (EFAH), which help the model better detect differently sized obstacles, in particular ones that are smaller or hidden (Wang et al. (2024)). It is also important to note that many object detection applications that navigate users towards their desired object are behind a paywall or not available for public use.

Another study applied YOLOv8 to help visually impaired users detect objects in real time on Android devices by providing auditory feedback (More et al. (2025)). Their application allowed users to be alerted to nearby objects. Our project hopes to build on this work by providing real-time auditory guidance to navigate around obstacles towards an user specified object. In addition, by running our system on a web platform instead of an application we hope to increase accessibility to be used across multiple devices.

YOLO, initially released in 2015, is a model available on Ultralytics used for CV tasks, including object detection and segmentation on live video (Ultralytics n.d.). The model has iteratively been improved until the most recent release, YOLO26. There are various YOLO models available for object tracking, and these models are trained either on the COCO or Open Images V7 dataset. The COCO (Common Objects in Context) dataset was released by Microsoft and provides 2,500,000 labeled objects within 328,000 images for 91 object classes (Lin et al.

2015). The Open Images V7 dataset contains around 9,000,000 images, 16,000,000 labeled objects, and 600 object classes ("Open Images V7" n.d., Kuznetsova et al. 2020).

YOLO26, trained on the COCO dataset, avoids non-maximum suppression, improves inference speed and demand, and introduces metrics to improve performance within various supported CV tasks (Ultralytics n.d.). YOLOv8, released in 2023, additionally provides a model pretrained on the Open Images V7 dataset (Ultralytics 2023). Further, the YOLO-World model builds from the YOLOv8 model by integrating user text to identify specified components and objects within a provided image (Ultralytics 2024).

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